



Sum of Fibonacci Numbers | Set 2

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Given a positive number **N**, the task is to [find the sum of the first \(N + 1\) Fibonacci Numbers](#).

Examples:

Input: $N = 3$

Output: 4

Explanation:

The first 4 terms of the Fibonacci Series is {0, 1, 1, 2}. Therefore, the required sum = $0 + 1 + 1 + 2 = 4$.

Input: $N = 4$

Output: 7

Naive Approach: For the simplest approach to solve the problem, refer to the [previous post](#) of this article.

Time Complexity: $O(N)$

Auxiliary Space: $O(1)$

Efficient Approach: The above approach can be optimized by the following observations and calculations:

Let **S(N)** represent the sum of the first **N** terms of Fibonacci Series. Now, in order to simply find **S(N)**, [calculate the \(N + 2\)th Fibonacci number](#) and subtract **1** from the result. The **Nth term** of this series can be calculated by the formula:

$$F_N = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^N}{\sqrt{5}}$$

Now the value of **S(N)** can be calculated by **(F_{N+2} - 1)**.

Therefore, the idea is to calculate the value of **S_N** using the above formula:

Below is the implementation of the above approach:

C++ Java Python3 C# JavaScript

```
// C++ program for the above approach
#include <bits/stdc++.h>
using namespace std;

// Function to find the sum of
// first N + 1 fibonacci numbers
```

```

void sumFib(int N)
{
    // Apply the formula
    long num = (long)round(pow((sqrt(5) + 1)
                               / 2.0, N + 2)
                       / sqrt(5));

    // Print the result
    cout << (num - 1);
}

// Driver Code
int main()
{
    int N = 3;
    sumFib(N);
    return 0;
}

// This code is contributed by Dharanendra L V.

```

Output:

4

Time Complexity: $O(\log n)$

Auxiliary Space: $O(1)$

Alternate Approach: Follow the steps below to solve the problem:

- The **Nth Fibonacci number** can also be calculated using the [generating function](#).
- The generating function for the **Nth Fibonacci number** is given by:

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$$G(X) = \frac{X}{(1-X-X^2)}$$

- Therefore, the generating function of the sum of Fibonacci numbers is given by:

$$G'(X) = \frac{X}{((1-X-X^2)*(1-X))}$$

- After partial fraction decomposition of **G'(X)**, the value of G'(X) is given by:

$$G'(X) = \frac{a}{(a+1)*(a-b)*(X+a)} + \frac{b}{(b+1)*(b-a)*(X+b)} + \frac{1}{(a+1)*(b+1)*(X-1)}$$

where,

$$a = \frac{(1+\sqrt{5})}{2}, b = \frac{(1-\sqrt{5})}{2}$$

- Therefore, the formula for the sum of the Fibonacci numbers is given by:

$$S_N = \left\lfloor \frac{1}{b^{N+1} + b^N + b^{N-1} + b^{N-2}} \right\rfloor - 1$$

Below is the implementation of the above approach:

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```
using namespace std;

// Function to find the sum of
// first N + 1 fibonacci numbers
void sumFib(int N)
{
    // Apply the formula
    double num = (1 - sqrt(5)) / 2;

    long val = round(
        abs(1
            / (pow(num, N + 2)
              + pow(num, N + 1)
              + pow(num, N)
              + pow(num, N - 1)))
        - 1);

    // Print the result
    cout<<val;
}

// Driver Code
int main()
{
    int N = 3;
```

```
// Function Call
sumFib(N);
}

// This code is contributed by 29AjayKumar
```

Output:

4

Time Complexity: $O(\log n)$
Auxiliary Space: $O(1)$

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